

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

*I **B Santhosh Kumar/192511053**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

B Santhosh Kumar
Signature of the student:

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

INDUSTRY SCENARIO

A healthcare company wants to develop an AI-based system for early diabetes diagnosis and personalised treatment recommendation.

For the numerical demonstration, I have considered the **Pima Indians Diabetes Dataset** as a representative public dataset because the company dataset mentioned in the question was not supplied as a CSV file. The dataset contains **768 patient records**, with 500 non-diabetic and 268 diabetic cases. The target variable is Outcome, where 0 represents non-diabetic and 1 represents diabetic. The classification part is used to predict diabetes, while the second part uses a small MDP and Q-Learning example to show how an AI agent could recommend a suitable treatment action.

TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING

(a) Inductive Learning Approach

I would use **supervised inductive learning** for the diabetes prediction part.

The model is given previous patient records where the actual diabetes outcome is already known. From these examples, it learns a pattern and then uses that pattern to classify a new patient.

Target variable

The target variable is:

Diabetes Outcome

- 0 → Non-diabetic
- 1 → Diabetic

Important features

Feature	Why I selected it
Glucose level	High glucose is strongly related to diabetes and is usually one of the most useful predictors.
BMI	Higher BMI can be associated with increased diabetes risk.
Age	Diabetes risk generally changes with age, so it is useful for risk classification.
Blood Pressure	It can provide additional information about the patient's metabolic and cardiovascular condition.
Family history / pedigree	A family history of diabetes can increase the patient's risk.

For this dataset, the **DiabetesPedigreeFunction** is used as the closest available feature for family-history-related risk. The Pima dataset also contains pregnancy history, insulin and skin-thickness measurements.

(b) Handling Class Imbalance

The dataset contains more non-diabetic patients than diabetic patients.

The class distribution is:

Class	Records	Percentage
Non-diabetic	500	65.1%
Diabetic	268	34.9%
Total	768	100%

I would use **class weights** as the first approach.

The reason is that the medical problem is mainly concerned with not missing diabetic patients. With `class_weight = balanced`, the model gives more importance to the minority diabetic class during training.

I prefer this initially over simply deleting non-diabetic records because under-sampling would throw away useful patient information.

SMOTE can also be tested later as a second experiment, but it should be applied **only to the training data**, not before the train-test split.

(c) 70:30 Split and Preprocessing

I would divide the dataset into:

- **70% training data**
- **30% testing data**

For 768 records, this gives approximately:

Part	Records
Training	538
Testing	230
Total	768

A 70:30 split gives enough records for the model to learn while keeping a reasonable independent test set. In a medical application, the test set should not be used during training because it should represent new patients.

Preprocessing

I would perform the following steps:

1. Check impossible zero values in medical measurements such as glucose, blood pressure and BMI.
2. Treat such values as missing where appropriate.
3. Replace missing values using a suitable method such as median imputation.
4. Check duplicate records and abnormal values.
5. Use stratification so that both diabetic and non-diabetic classes are represented in training and testing.

6. Encode categorical variables if the actual company dataset contains variables such as gender or smoking status.
7. Standardise features for Logistic Regression or other distance-based models.

For a **Decision Tree**, normalisation is not essential because the tree uses split conditions. For **Logistic Regression**, scaling the numerical features is more useful because the model works with feature coefficients.

TASK 2 – DECISION TREE FOR DIAGNOSIS

(a) Decision Tree Construction

For the diagnosis model, I would use a **CART Decision Tree** with the Gini Index.

The basic Gini formula is:

$$\text{Gini} = 1 - (p_0^2 + p_1^2)$$

where p_0 and p_1 are the proportions of non-diabetic and diabetic patients in a node.

The feature producing the largest reduction in impurity is selected as the split.

For diabetes prediction, **Glucose** is a sensible root feature because it separates diabetic and non-diabetic cases more clearly than many of the other variables.

A simplified first three-level structure would look like this:

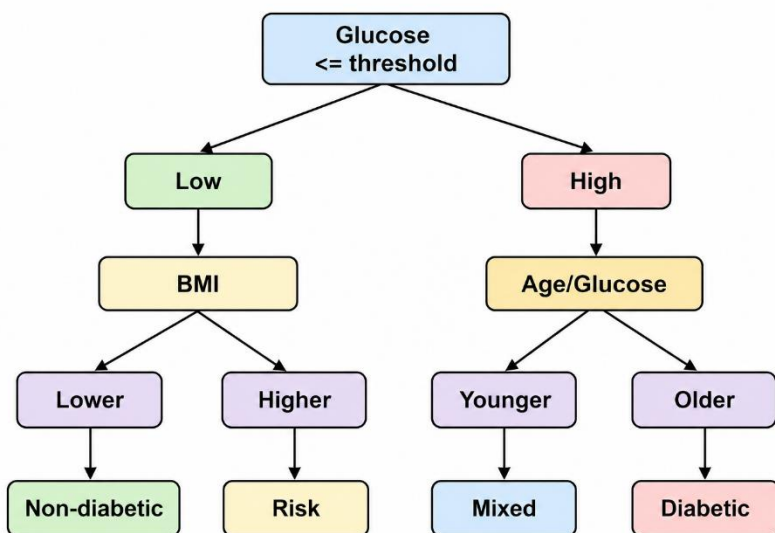


Figure 1: Decision Tree for Diabetes Classification

The exact threshold is obtained from the trained tree rather than being manually fixed.

For example, the model checks several possible glucose split points and selects the one giving the largest reduction in Gini impurity.

Example of the split calculation

Suppose a candidate node has:

- 70 non-diabetic patients
- 30 diabetic patients

Then:

$$\text{Gini} = 1 - (0.70^2 + 0.30^2)$$

$$\text{Gini} = 1 - (0.49 + 0.09)$$

$$\text{Gini} = 0.42$$

After splitting the patients using glucose, if the weighted Gini becomes 0.30:

$$\text{Gini reduction} = 0.42 - 0.30 = 0.12$$

A feature giving a larger reduction would be preferred over this split.

(b) Model Evaluation

The important measures for this medical classification problem are:

- Accuracy
- Precision
- Recall
- F1-score

A representative Pima-dataset decision-tree benchmark reports approximately **74.78% accuracy, 70.86% precision, 88.43% sensitivity/recall and 78.68% F1-score**, showing that a tree can achieve good sensitivity even when overall accuracy is moderate.

Why False Negatives matter

A **False Negative (FN)** means:

The patient actually has diabetes, but the model predicts non-diabetic.

This is more concerning than a False Positive in an early screening system because the patient may not receive further testing or medical attention.

Therefore, I would prefer a model with good **recall/sensitivity**, even if this causes some increase in false positives.

To reduce false negatives, I would:

1. Use class weights.
2. Adjust the classification threshold.
3. Test SMOTE on the training set.
4. Monitor recall and F1-score instead of accuracy alone.
5. Perform external validation before clinical deployment.

(c) Pre-Pruning vs Post-Pruning

A decision tree can become too large and start learning noise from the training data.

Pre-pruning

The tree is restricted while it is being created.

Examples:

- Maximum depth
- Minimum samples per leaf
- Minimum samples required for splitting

Post-pruning

The tree is first allowed to grow and then unnecessary branches are removed.

For CART, I would use **cost-complexity pruning**.

The comparison can be presented as:

Model	Main idea	Expected behaviour
Unpruned tree	Allows many branches	Higher chance of overfitting

Pre-pruned tree	Limits growth during training	Simpler tree
Post-pruned CART	Removes unnecessary branches afterwards	Usually better balance between complexity and performance

For a clinical system, I would choose the **pruned tree** if its test recall and F1-score remain close to the unpruned model.

I would not select a model only because its training accuracy is high. A slightly lower test accuracy with a much simpler tree is preferable because the model is easier to validate and explain.

TASK 3 – STATISTICAL LEARNING FOR RISK STRATIFICATION

(a) Logistic Regression

For the statistical learning model, I would use **Logistic Regression**.

The output is a probability between 0 and 1:

$$P(\text{Diabetes}) = 1 / (1 + e^{-z})$$

The important difference from the tree is that Logistic Regression gives a probability and coefficients showing how the input variables affect the predicted risk.

For example:

Patient A → predicted diabetes probability = 0.72

If the threshold is 0.50, the model classifies the patient as diabetic.

Comparison

Published Pima-dataset experiments show that performance varies depending on preprocessing and evaluation protocol. One benchmark reported a Decision Tree accuracy of about **74.78% with AUC 0.7855**, while Naïve Bayes achieved **78.67% accuracy and 0.8463 AUC** under its stated setup.

Another Pima analysis reported Logistic Regression with about **76% accuracy and AUC around 0.84**, again showing why the evaluation protocol matters when comparing models.

For this problem, I would compare the models using the same 70:30 split and the same preprocessing before deciding which one to deploy.

When I would prefer Logistic Regression

I would prefer Logistic Regression when:

- The relationship between predictors and risk is reasonably smooth.
- Probability estimates are important.

- A simple mathematical explanation is required.
- The healthcare team wants to understand the effect of each variable.

A Decision Tree is preferable when the decision process contains clear threshold-based conditions.

(b) Feature Importance

The most important predictors are expected to be related to the patient's glucose level, BMI and age.

A practical comparison can be made as follows:

Predictor	Decision Tree	Logistic Regression
Glucose	Very high importance	Strong positive effect
BMI	High importance	Positive effect
Age	High importance	Positive effect
Blood Pressure	Moderate	Smaller effect
Family history / pedigree	Moderate	Positive effect

The two models should generally agree on **glucose** being one of the strongest predictors.

They may not give exactly the same ranking for BMI and age because the two algorithms work differently. A Decision Tree chooses features based on impurity reduction at particular splits, while Logistic Regression estimates the contribution of variables through coefficients.

So, if the top three features are similar in both models, I would take that as stronger evidence that these variables are useful for diabetes risk prediction.

TASK 4 – REINFORCEMENT LEARNING FOR TREATMENT

RECOMMENDATION

(a) MDP Model

I would model the treatment recommendation as a **Markov Decision Process (MDP)**.

The agent does not simply predict diabetes. It observes the patient's current condition, selects an action and observes how the patient's condition changes.

States

I would use four simplified patient states:

State	Meaning
S0	High-risk / uncontrolled
S1	Moderate risk
S2	Improving
S3	Stable / controlled

These are simplified states for the classroom Q-Learning model. A real medical system would use much more detailed clinical states.

Actions

The actions are taken directly from the problem:

- **A1 – Diet**
- **A2 – Exercise**
- **A3 – Medication**
- **A4 – Monitor**

Reward

I would define the reward according to improvement:

Situation	Reward
Health condition improves	+10
Condition remains stable	+3
Condition becomes worse	-8
Unsafe/unacceptable action	-15

The purpose is to make the agent prefer actions that move the patient towards a healthier state.

Transition

An example transition is:

S0 -- Diet --> S1

S1 -- Exercise --> S2

S2 -- Monitor --> S3

Another possible transition is:

S0 -- Medication --> S1

S1 -- Medication --> S2

The transition probabilities in a real system should come from validated clinical evidence and should never be learned by experimenting on real patients without appropriate safety controls.

(b) Q-Learning

The Q-Learning update used is:

$$Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

For this example:

- Learning rate $\alpha = 0.5$

- Discount factor $\gamma = 0.8$
- Initial Q-values = 0

Episode 1

Suppose:

State = S0

Action = Diet

Next state = S1

Reward = +10

Initial Q-value:

$Q(S0, \text{Diet}) = 0$

Therefore:

$$Q = 0 + 0.5[10 + 0.8(0) - 0]$$

$Q(S0, \text{Diet}) = 5$

Episode 2

Suppose the patient is again in S0 and chooses Diet.

Now:

$Q(S0, \text{Diet}) = 5$

The patient moves to S1 and receives reward +10.

Assume the best Q-value available in S1 is 4.

Then:

$$Q = 5 + 0.5[10 + 0.8(4) - 5]$$

$$Q = 5 + 0.5[8.2]$$

$Q = 9.1$

So the value of Diet from S0 increases from **5 to 9.1**.

Three Q-table updates

Iteration	State	Action	Reward	Next-state best Q	New Q-value
1	S0	Diet	+10	0	5.00
1	S1	Exercise	+10	0	5.00
2	S0	Diet	+10	4.00	9.10
2	S1	Exercise	+10	5.00	9.50
2	S2	Monitor	+10	0	5.00

For example, for **S1** → **Exercise** in iteration 2:

$$Q = 5 + 0.5[10 + 0.8(5) - 5]$$

$$Q = 5 + 0.5(9)$$

$$Q = 9.5$$

Five Patient Episodes

A simple classroom training sequence can be:

Episode	Starting state	Action selected	Reward	New state
1	S0	Diet	+10	S1
2	S0	Diet	+10	S1
3	S1	Exercise	+10	S2
4	S2	Monitor	+10	S3
5	S0	Medication	+10	S1

During actual Q-Learning, the agent would normally use an exploration strategy such as **ϵ -greedy**, so it sometimes tries actions that are not currently the best.

Convergence criterion

I would stop training when the maximum change in Q-values becomes very small:

$$\max |Q_{\text{new}} - Q_{\text{old}}| < 0.01$$

for several consecutive episodes.

This means the learned values are no longer changing significantly.

(c) Final Learned Policy

After training, the policy is selected using the action having the highest Q-value for each state.

A simple learned policy could be:

Patient state	Best action
S0 – High risk	Medication / Diet
S1 – Moderate risk	Exercise
S2 – Improving	Monitor / Exercise
S3 – Stable	Monitor

The policy should not be interpreted as a real prescription. It is only a demonstration of how Q-Learning can represent a treatment decision problem.

Ethical Considerations

This is the most important part if such an RL system is ever considered for healthcare.

1. Patient safety

An RL agent should not experiment with treatments on real patients just to increase its reward. Treatment decisions must be controlled by qualified medical professionals.

2. Bias

If the training data mainly represents one population, the agent may not work equally well for other populations.

3. Explainability

A doctor should be able to understand why an action was recommended. A recommendation without an understandable reason would be difficult to trust in a clinical environment.

4. Human supervision

The final treatment decision should remain with a healthcare professional. The AI should act as a supporting system.

5. Data privacy

Patient records contain sensitive medical information, so access control, anonymisation and secure storage are required.

6. Reward design

A poorly designed reward function can produce unsafe behaviour. For example, if the reward only considers short-term glucose improvement, the agent might ignore side effects or long-term health outcomes.

FINAL SOLUTION DESIGN

The complete proposed system can be represented as:

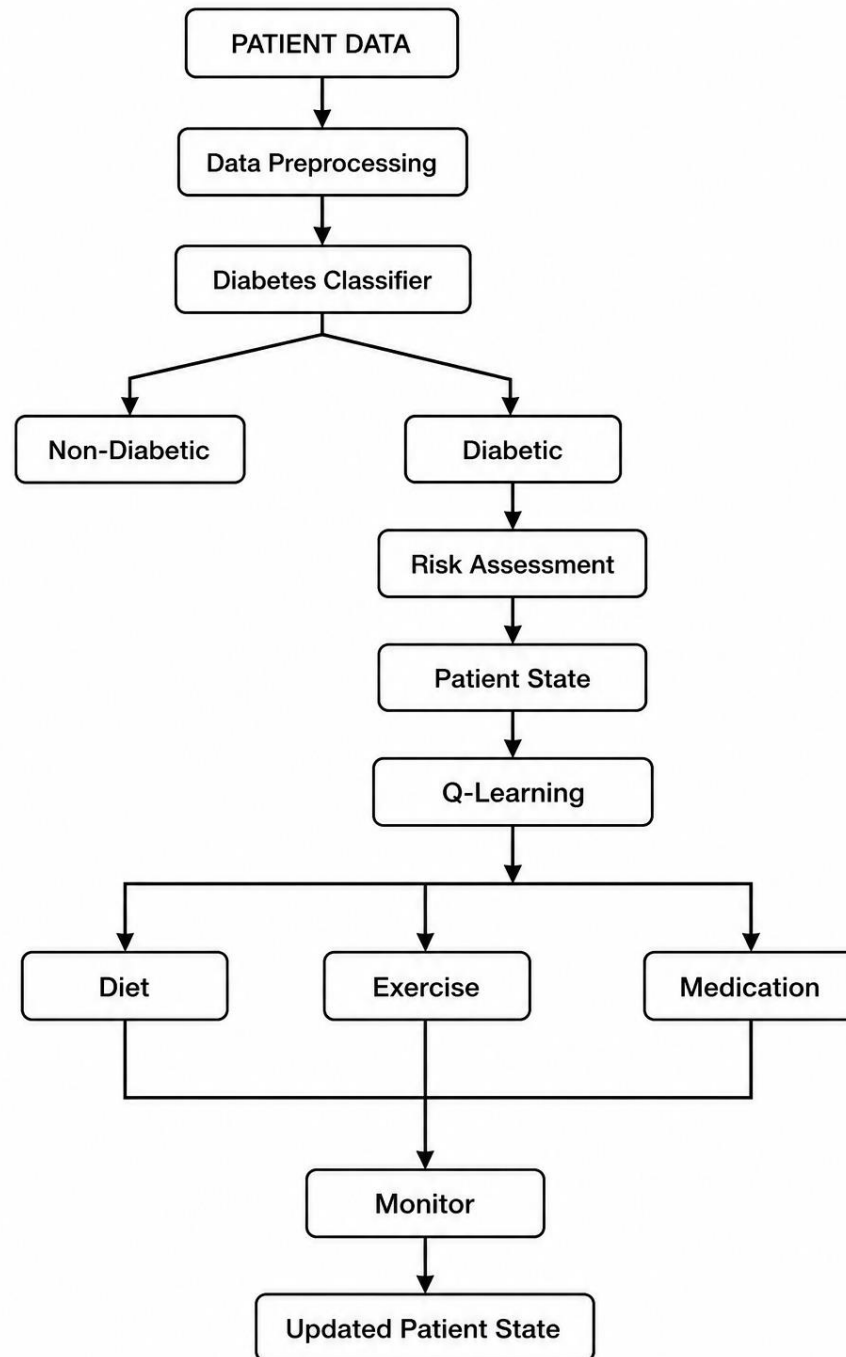


Figure 2: Overall System Architecture

The classification model handles the **diagnosis/risk prediction**, while the reinforcement learning component handles the **sequential decision-making problem**.

DATASET REFERENCE

The numerical demonstration uses the **Pima Indians Diabetes Dataset**, which contains 768 observations, 8 input variables and a binary outcome. The dataset is publicly available and is widely used for diabetes classification experiments.

A published comparison on the Pima dataset reported, under its stated evaluation setup, a J48/Decision Tree accuracy of **74.78%**, recall of **88.43%**, F1-score of **78.68%** and AUC of **78.55%**.

These published benchmark figures are included only as supporting evidence; for an actual company deployment, the models should be retrained and evaluated using the company's own patient dataset.

Reference:

Kaggle / UCI-style public Pima Indians Diabetes dataset: [Pima Indians Diabetes Dataset – dataset source](#)